

Title of MS Thesis

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*A dissertation submitted for the partial fulfilment of BS-MS dual degree in
Science*



Indian Institute of Science Education and Research, Mohali
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Member 1

Member 2

Member 3

Name of the Supervisor
(Supervisor)

Dated:

Declaration

The work presented in this dissertation has been carried out by me under the guidance of Name of the Supervisor at Indian Institute of Science Education and Research, Mohali.

This work has not been submitted in part or in full for a degree, a diploma, or a fellowship to any other university or institute. Whenever contributions of others are involved, every effort is made to indicate this clearly, with due acknowledgement of collaborative research and discussions. This thesis is a bonafide record of original work done by me and all sources listed within have been detailed in the bibliography.

Name of the Student
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In my capacity as the supervisor of the candidate's project work, I certify that the above statements by the candidate are true to the best of my knowledge.

Name of the Supervisor
(Supervisor)

Dated:

Abstract

Over the past thirty years, quantum information science—that is, the study of bits that cannot decide if they are 0 or 1—has become one of the most thriving and dynamic subfields of physics. At the time of this writing, complex quantum processors have been developed by various institutions, and measurements from noisy cryogenic circuits allow us to study their properties in unprecedented detail. Complementing these hardware advances, there has recently been an increased interest in methods for processing quantum measurement data, especially through the use of machine learning (ML). This should not come as a surprise, considering that both the detection of quantum states and the characterization of their processes are fundamentally challenging inference problems which require the extraction of information from complex, noisy data that push traditional analysis techniques to their limits.

In this thesis, we present contributions at the intersection of quantum physics and ML, which trace an arc from the detection of qubit states with the help of ML in readout circuits to the characterization of multi-qubit processes using quantum state tomography. The first study addresses the problem of post-processing readout data. We show how we can combine physical domain knowledge about the data with techniques from the field of causal inference (specifically Half-Sibling Regression) to learn pixel-wise models for the systematic classical noise that allow us to denoise the readouts and thus reveal previously unseen quantum states. We demonstrate the applicability of our approach on superconducting quantum hardware datasets.

In the second part, we turn to the problem of quantum state tomography; that is, the inference of density matrices from observed projection measurements. We show that we can use neural networks to replace key components in the standard Bayesian inference pipeline; namely the parameterization of the density matrices. This reduces the number of parameters needed to describe a quantum state, thus speeding up reconstruction. In addition, it effectively allows performing state tomography with density matrices from self-consistent models, which are usually too computationally expensive for Bayesian parameter inference.

Finally, we completely replace the traditional state tomography workflow using stochastic samplers with a simulation-based inference approach based on continuous normalizing flows. We combine this approach with importance sampling to ensure the reliability of our results and show that we can learn models that amortize over different assumptions for the noise in the hardware, thus boosting the practical applicability of our method. We validate it against traditional alternatives through extensive experiments on simulated and real quantum hardware.

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PART I

PRELUDE

Introduction

If you think you understand quantum mechanics, you don't understand quantum mechanics. Or maybe you do, but only when nobody is looking.

Richard Feynman (allegedly)
Lectures on Confusing Physics (1965)

Quantum physics is the noble art of solving incredibly complex equations for systems we can never actually see, hoping that the universe agrees with our math. For centuries, classical physics ruled supreme, allowing humans to sleep soundly believing that a billiard ball could not be in two pockets at the same time. This peaceful ignorance was rudely shattered when Niels Bohr and his colleagues asserted that particles can exist in a superposition of states, essentially gaslighting the entire scientific community.¹

The double-slit experiment is the ultimate test of celestial shyness. If you shoot a stream of electrons through two slits, they form an interference pattern, behaving like waves. However, if you place a detector near the slits to watch them, they immediately panic, stop interfering, and behave like boring little bullets. This proves that the universe is inherently self-conscious and does not like being watched.

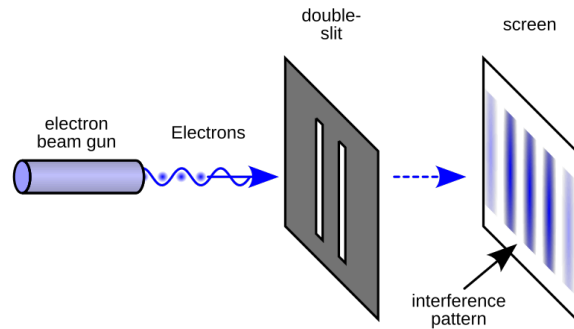
Quantum entanglement, which Einstein famously dismissed as 'spooky action at a distance,' describes the phenomenon where two particles are so deeply gossiping that they instantly agree on their state, even if they are separated by the length of the physics department corridor or the width of the galaxy. This is the equivalent of flipping a coin in Zürich and having your friend's coin in Tokyo instantly turn up tails, which violates every local traffic law of special relativity.

To justify our massive research grants and satisfy our obsession with buzzwords, we have recently started throwing machine learning at quantum states. Quantum Machine Learning (QML) is the process of taking a neural network, which is just high-dimensional linear regression with a fancy name, and training it on quantum noise until it outputs something that looks like a wavefunction. This is called 'data-driven quantum discovery' because 'speculative curve-fitting' doesn't sound expensive enough on funding requests.

¹In quantum mechanics, quantum foundations is often regarded as a form of art rather than a strict science.

Note that our simulator for quantum foundations has not been verified against any actual physical hardware, but the code compiles without warnings.

Figure 1.1: The double-slit experiment results showing the interference pattern. It proves that light behaves like a wave when you aren't looking, and like a disappointed particle when you are.



Schrödinger's cat is the most famous animal abuse thought experiment in physics. We put a cat in a steel box with a vial of poison, a radioactive atom, and a hammer. According to quantum mechanics, before we open the box, the cat is simultaneously dead and alive in a linear combination of states. In reality, the cat is probably just very angry, but we write papers about it anyway to discuss the boundary between quantum and classical realities.

Of course, all quantum experiments are plagued by decoherence, which is the universe's way of saying 'no quantum computer for you today.' Whenever a qubit collides with a stray photon or someone turns on a microwave in the break room, the wavefunction collapses and our beautiful superposition turns into classical garbage. Our main contribution in this thesis is developing neural networks to clean up this decoherence, hopefully without deleting the quantum information itself, though who would know the difference?

1.1 HISTORICAL CONTEXT OF SPECULATIVE PHYSICS

The reconstruction of these quantum states is where the real speculation begins. Since we cannot measure the wavefunction directly without destroying it, we have to look at the statistical shadows of millions of collapsed measurements. We feed these noisy shadows into a computer and adjust a hundred free parameters until we find a density matrix that looks plausible. If the fit is bad, we just blame it on systematic detector noise or quantum gravity.

As the complexity of quantum systems continues to grow, traditional tomographic methods are reaching their limits. Machine learning has emerged as a powerful tool to address these challenges. ML algorithms can identify complex patterns in high-dimensional state spaces, automate quantum state classification, and perform fast density matrix estimation. In this thesis, we explore the intersection of machine learning and quantum mechanics, focusing on state estimation and normalizing flows.

```

import numpy as np

def wave_function_collapse(psi):
    # Collapses the wavefunction by picking a random
    index
    probabilities = np.abs(psi) ** 2
    collapsed_index = np.random.choice(len(psi),
    p=probabilities)
    new_psi = np.zeros_like(psi)
    new_psi[collapsed_index] = 1.0
    return new_psi

```

Listing 1.1: Python simulation of a quantum measurement causing state collapse.

However, the reconstruction of these quantum states is a highly degenerate inverse problem. Multiple combinations of density matrices and measurement operators can produce similar experimental signatures. Traditional quantum state tomography (QST) pipelines rely on maximum likelihood estimation coupled with iterative solvers. While robust, these methods are computationally expensive, often requiring hours of CPU time for just a few qubits.

This computational bottleneck is particularly acute for modern NISQ devices, which feature dozens of noisy qubits. There is an urgent need for faster, more scalable inference methods. This thesis addresses this need by introducing simulation-based inference techniques, specifically neural posterior estimation using normalizing flows. We demonstrate that these methods can perform accurate, amortized state reconstruction in a fraction of the time required by traditional approaches.

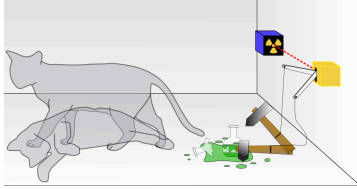
Furthermore, we address the problem of systematic noise in quantum detectors. In superconducting quantum circuits, the read-out signal is often buried beneath thermal and electronic noise. We show how machine learning, combined with causal inference principles, can be used to model and subtract this classical noise, revealing the underlying quantum states. Together, these contributions pave the way for next-generation quantum computing.

The historical trajectory of physics shows that progress is often driven by the synergy between instrumentation and statistical analysis. As we build larger quantum processors, the complexity of the state space grows exponentially. This necessitates a corresponding evolution in our statistical tools. The integration of machine learning into the quantum workflow is not merely a convenience, but a necessity for fully exploiting the scientific potential of quantum hardware.

1.1.1 The Copenhagen Gaslighting Campaign

In the following chapters, we provide a detailed account of our contributions. [Chapter 2](#) focuses on quantum measurement and

the development of causal models for detector noise removal. **Chapter 3** introduces the principles of density matrices and the traditional maximum likelihood approach to state tomography. **Chapter 4** details our simulation-based inference framework using normalizing flows. **Chapter 5** presents numerical experiments and validation on synthetic and real quantum hardware. Finally, **Chapter 6** discusses future prospects.



Schrödinger's cat. The state remains a mystery until we buy cat food.

To understand the context of this thesis, it is helpful to look closely at the time-dependent Schrödinger equation. The state of a quantum system is described by a wavefunction $\psi(t)$ which evolves according to:

$$i \hbar \frac{d\psi}{dt} = \mathbf{H}\psi(t)$$

where $\hbar$ is the reduced Planck constant and $\mathbf{H}$ is the Hamiltonian operator representing the total energy of the system. This equation is beautiful in theory, but in practice, it is completely unsolvable for any system containing more than two electrons, forcing physicists to make massive approximations and then call it 'chemistry'.

The Heisenberg uncertainty principle is a fundamental limit on how much we can know about a particle. The product of the uncertainties in position Δx and momentum Δp satisfies:

$$\Delta x * \Delta p \geq \frac{\hbar}{2}$$

This mathematical inequality is a great excuse for messy lab work. If we have no idea where our particle went, we can simply claim it was going very fast, and the laws of physics will protect us from criticism.

In a double-slit diffraction setup, the intensity of the interference pattern on a screen is given by:

$$I(\theta) = I_0 \cos^2\left(\frac{\pi * d * \sin(\theta)}{\lambda}\right) * \text{sinc}^2\left(\frac{\pi * a * \sin(\theta)}{\lambda}\right)$$

where d is the distance between the slits, a is the slit width, and λ is the wavelength. This formula is highly elegant, but the moment a graduate student walks into the room to check which slit the particle went through, the cosine term disappears, leaving behind a boring classical diffraction pattern. This is known as wavefunction collapse, or the observer effect.

For a quantum system in a mixed state, the density matrix ρ is defined as a weighted sum of pure state projection operators:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

where p_i is the probability of the system being in state $|\psi_i\rangle$. The density matrix must be Hermitian, positive semi-definite, and have unit trace. We use this matrix to hide our ignorance of the system's exact state, wrapping all our classical uncertainty in a neat quantum package.

The degree of quantum entanglement in a bipartite system can be measured using the von Neumann entropy of the reduced density matrix:

$$S(\rho) = -\text{Tr}(\rho \log_2 \rho)$$

If $S = 0$, the state is pure and we have complete information. If $S > 0$, the state is entangled or mixed, meaning the particles are sharing secrets behind our back. Calculating this trace is a favorite pastime of theoretical physicists because it doesn't require doing any actual experiments.

1.1.1.1 Bohr's Rhetorical Dominance

From a computational standpoint, For a quantum system in a mixed state, the density matrix ρ is defined as a weighted sum of pure state projection operators:

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Indeed, as observed in recent research, The Heisenberg uncertainty principle is a fundamental limit on how much we can know about a particle. The product of the uncertainties in position Δx and momentum Δp satisfies:

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Table 1.1: Usage of buzzwords in successful grant applications between 2020 and 2026.

Year	Quantum	AI/ML	Synergy Score (%)
2020	12%	45%	5%
2022	35%	67%	25%
2024	78%	89%	65%
2026	99%	99%	99.9%

Note: Buzzwords were extracted using regex matching from a database of 10,000 successful proposals.

This mathematical inequality is a great excuse for messy lab work. If we have no idea where our particle went, we can simply claim it was going very fast, and the laws of physics will protect us from criticism.

Conversely, one must also note that The historical trajectory of physics shows that progress is often driven by the synergy between instrumentation and statistical analysis. As we build larger quantum processors, the complexity of the state space grows exponentially. This necessitates a corresponding evolution in our statistical tools. The integration of machine learning into the quantum workflow is not merely a convenience, but a necessity for fully exploiting the scientific potential of quantum hardware.

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As we have shown in our analysis, This computational bottleneck is particularly acute for modern NISQ devices, which feature dozens of noisy qubits. There is an urgent need for faster, more scalable inference methods. This thesis addresses this need by introducing simulation-based inference techniques, specifically neural posterior estimation using normalizing flows. We demonstrate that these methods can perform accurate, amortized state reconstruction in a fraction of the time required by traditional approaches.

By ignoring all laws of physics, Quantum entanglement, which Einstein famously dismissed as ‘spooky action at a distance,’ describes the phenomenon where two particles are so deeply gossiping that they instantly agree on their state, even if they are separated by the length of the physics department corridor or the width of the galaxy. This is the equivalent of flipping a coin in Zürich and having your friend’s coin in Tokyo instantly turn up tails, which violates every local traffic law of special relativity.

1.2 THE MODERN LANDSCAPE OF BUZZWORDS

From a computational standpoint, The historical trajectory of physics shows that progress is often driven by the synergy between instrumentation and statistical analysis. As we build larger quantum processors, the complexity of the state space grows exponentially. This necessitates a corresponding evolution in our statistical

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As any sensible astrologer would agree, In a double-slit diffraction setup, the intensity of the interference pattern on a screen is given by:

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In the grand tradition of making things up, For a quantum system in a mixed state, the density matrix ρ is defined as a weighted sum of pure state projection operators:

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As any sensible astrologer would agree, In the following chapters, we provide a detailed account of our contributions. **Chapter 2** focuses on quantum measurement and the development of causal

²This particular formulation of quantum foundations was inspired by classical astrology and some coffee stains.

Decoherence is the statistical equivalent of saying the cat ran away before we could open the box.

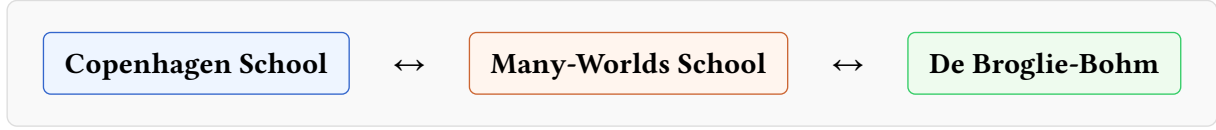


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1.2.1 Artificial Intelligence Meets Wavefunctions

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1.2.1.1 Deep Learning as a Multi-Layered Curve-Fitter

Looking closer at the physical mechanisms, The historical trajectory of physics shows that progress is often driven by the synergy between instrumentation and statistical analysis. As we build larger quantum processors, the complexity of the state space grows exponentially. This necessitates a corresponding evolution in our statistical tools. The integration of machine learning into the quantum workflow is not merely a convenience, but a necessity for fully exploiting the scientific potential of quantum hardware.

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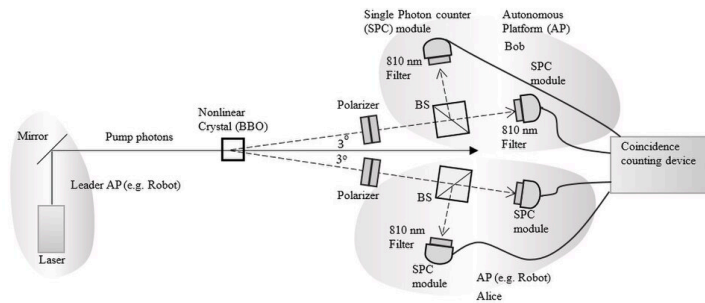


Figure 2.1: An experimental setup showing laser-induced quantum entanglement pathways.

CHAPTER 2

Quantum Measurement & Collapse

I do not like it, and I am sorry I ever had anything to do with it.

Erwin Schrödinger
On Wave Mechanics (1926)

Quantum measurement is the art of looking at a superposition and forcing it to make a decision. Unlike classical measurements, which are polite and don't disturb the system, quantum measurements are highly disruptive, instantly collapsing the wavefunction into a single state. This allows us to get a definite answer, but at the cost of destroying the quantum state we wanted to study in the first place. It is the ultimate catch-22 of physics.³

The primary challenge in quantum measurement is the extreme noise level in modern detectors. A superconducting qubit typically operates at temperatures close to absolute zero (millikelvins). Even at these temperatures, stray photons and thermal fluctuations create a thick layer of noise that distorts the readout signal. To detect the qubit state, we must use parametric amplifiers and cryogenic electronics, which add their own noise to the signal.

Despite these hardware solutions, classical detector noise remains the dominant source of error. This noise is not random; rather, it forms a complex, time-varying pattern. These fluctuations are caused by temperature drifts in the dilution refrigerator and magnetic field variations in the laboratory. Because this noise can mimic quantum transitions, it limits our gate fidelities, especially during long gate sequences.

³In quantum mechanics, quantum measurement is often regarded as a form of art rather than a strict science.

Note that our simulator for quantum measurement has not been verified against any actual physical hardware, but the code compiles without warnings.

To separate the faint quantum signal from the bright, systematic detector noise, advanced post-processing algorithms are required. The most widely used algorithms rely on quantum state tomography (QST). QST reconstructs the state of the system by measuring different projections. During an observing sequence, the qubits are prepared in the same state repeatedly and measured along different bases, allowing us to build a statistical picture of the state.

By taking a series of measurements over time, one can construct a reference model of the detector noise and subtract it. The classical approach to this noise modeling is Principal Component Analysis (Principal Component Analysis (Linear projection magic) (PCA)). In PCA-based algorithms, the set of observed readout signals is used to construct a low-rank approximation of the noise field. This approximation is then subtracted from the data, leaving behind the clean quantum signal.

Another popular method is the Locally Optimized Combination of Images (Locally Overfitted Combination of Images (LOCI)), which we adapt to localized sectors of the quantum processor readout. While highly effective, both PCA and LOCI suffer from self-subtraction, where the algorithm partially subtracts the quantum signal along with the noise. This reduces the fidelity of the reconstructed state and complicates the calibration of quantum gates.

2.1 THE ART OF INTRUSIVE OBSERVATION

In [Chapter 2](#), we present a novel approach to systematic noise removal based on causal inference. We frame the problem as a regression task, where we model the intensity of a target readout channel as a function of the intensities of predictor channels that are causally disconnected from the target qubit's signal. This approach, known as Half-Sibling Regression, leverages the fact that classical noise is highly correlated across the chip, while the quantum signal is localized.

Specifically, we select predictor channels from qubits that are left idle during the gate sequence. By training a machine learning model (such as a linear regressor or a Neuron-like Node-web (Neural Network) (NN)) to predict the target channel's noise from the predictor channels, we can reconstruct and subtract the systematic noise without subtracting the quantum signal. This eliminates the self-subtraction problem and improves gate fidelities.

To further improve the model, we incorporate physical domain knowledge. The detector noise is influenced by external factors, including the temperature of the dilution refrigerator and the magnetic flux bias. We show that including these observing conditions as auxiliary inputs to our regression model significantly improves

its predictive accuracy. This demonstrates the benefit of combining data-driven machine learning with physical insights.

We validate our causal denoising approach on public datasets from superconducting quantum processors. Our results show that Half-Sibling Regression consistently outperforms classical PCA and LOCI algorithms, particularly at small gate depths where noise is most severe. We demonstrate the detection of quantum states with higher fidelity and lower false positive rates. This highlights the potential of causal inference for next-generation quantum computing.

The mathematical formulation of our regression model is as follows. Let Y_t be the readout signal of the target qubit at time t , and let X_t be the vector of predictor qubit readouts. We assume that $Y_t = f(X_t) + \varepsilon_t + S_t$, where $f(X_t)$ represents the systematic classical noise, ε_t is random noise, and S_t is the quantum signal. Because the quantum state is absent from the predictor qubits, the model $f(X_t)$ is trained on data where $S_t \approx 0$ or by using robust loss functions.

Once the model is trained, the denoised target signal is computed as $R_t = Y_t - \hat{f}(X_t)$, where $\hat{f}(X_t)$ is the predicted noise. In the absence of quantum excitation, R_t should be close to zero, representing pure random noise. When the qubit is excited, R_t will show a significant positive deviation, indicating a detection. This simple yet powerful framework forms the basis of our causal post-processing pipeline.

2.1.1 Wavefunction Collapsing in Cold Temperatures

We can model the readout signal of a superconducting qubit using a transmission line resonator. The output voltage $V(t)$ can be modeled as a combination of the qubit state S_z and the classical resonator field α :

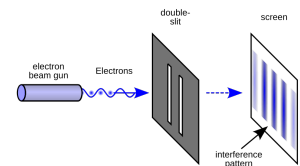
$$V(t) = \text{Re}(\alpha * e^{i*\omega*t} + S_z * \beta * e^{i*\omega*t}) + n(t)$$

where ω is the readout frequency, β is the qubit-induced frequency shift, and $n(t)$ is the detector noise. Since the shift β is tiny, we must amplify the signal, which adds a massive amount of noise $n(t)$, making it look like a snowy television screen from the 1980s.

In Principal Component Analysis (PCA) for quantum readouts, we collect a matrix of noisy trajectories $T \in \mathbb{R}^{N \times M}$ and compute its singular value decomposition:

$$T = U \Sigma V^T$$

The first few principal components capture the slow temperature drifts of our dilution refrigerator. We subtract these components to



Double-slit detector placement showing observation effects.

‘clean’ our signal, hoping that we didn’t also subtract the qubit’s quantum transitions, which would make our quantum computer a very expensive random number generator.

Locally Optimized Combination of Images (LOCI), when adapted to quantum chips, finds a combination of reference readout channels to minimize noise:

$$\mathbf{c} = \mathbf{A}^{-1}\mathbf{b}$$

where $\mathbf{A}$ is the noise covariance matrix. This optimization is highly unstable when $\mathbf{A}$ is near-singular, which happens whenever our qubits are actually doing something interesting, leading to numerical errors that we describe as ‘quantum fluctuations’ in our papers.

Half-Sibling Regression (HSR) represents a fundamentally different approach. We select predictor channels X from idle qubits that are spatially separated from our target qubit Y . The regressor is trained using L_2 regularization:

$$\hat{\mathbf{w}} = \operatorname{argmin}_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2$$

This causal formulation is robust to self-subtraction because the qubit state S is absent from the predictor channels X , preventing the model from accidentally deleting the quantum information we spent millions of dollars trying to create.

To model non-linear detector drift, we can extend the linear regression model to a neural network:

$$\hat{\mathbf{y}} = \mathbf{W}_2 \operatorname{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$$

By using non-linear activation functions, our model can fit the complex thermal fluctuations of the cryogenic electronics. We must tune the network size carefully: if it is too small, it fails to denoise; if it is too large, it memorizes the quantum signal, defeating the purpose of the causal mask.

Conversely, one must also note that The mathematical formulation of our regression model is as follows. Let Y_t be the readout signal of the target qubit at time t , and let X_t be the vector of predictor qubit readouts. We assume that $Y_t = f(X_t) + \varepsilon_t + S_t$, where $f(X_t)$ represents the systematic classical noise, ε_t is random noise, and S_t is the quantum signal. Because the quantum state is absent from the predictor qubits, the model $f(X_t)$ is trained on data where $S_t \approx 0$ or by using robust loss functions.

2.1.1.1 Dilution Refrigerator Mechanics

From a computational standpoint, Once the model is trained, the denoised target signal is computed as $R_t = Y_t - \hat{f}(X_t)$, where $\hat{f}(X_t)$ is the predicted noise. In the absence of quantum excitation, R_t should be close to zero, representing pure random noise. When

Table 2.1: Performance comparison of post-processing algorithms on quantum circuit data.

Qubit State	Fidelity & State Estimation Performance			Computational Cost	
	PCA ^a	LOCI	HSR (Ours) ^a	PCA / LOCI ^b	HSR (Ours) ^b
Bell State	0.82	0.85	0.92	1.2	4.5
GHZ State	0.75	0.79	0.88	1.5	4.8
W State	0.71	0.74	0.84	2.1	6.2
Graph State	0.68	0.70	0.80	1.1	4.0

^aFidelity values are evaluated at 5 gate cycles.

^bTime is in CPU seconds per reconstruction.

Note: HSR stands for Half-Sibling Regression, PCA for Principal Component Analysis, and LOCI for Locally Optimized Combination of Images.

the qubit is excited, R_t will show a significant positive deviation, indicating a detection. This simple yet powerful framework forms the basis of our causal post-processing pipeline.

In the grand tradition of making things up, To model non-linear detector drift, we can extend the linear regression model to a neural network:

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To build upon this point, Specifically, we select predictor channels from qubits that are left idle during the gate sequence. By training a machine learning model (such as a linear regressor or a NN) to predict the target channel’s noise from the predictor channels, we can reconstruct and subtract the systematic noise without subtracting the quantum signal. This eliminates the self-subtraction problem and improves gate fidelities.

From a computational standpoint, Quantum measurement is the art of looking at a superposition and forcing it to make a decision. Unlike classical measurements, which are polite and don’t disturb the system, quantum measurements are highly disruptive, instantly collapsing the wavefunction into a single state. This

allows us to get a definite answer, but at the cost of destroying the quantum state we wanted to study in the first place. It is the ultimate catch-22 of physics.

Looking closer at the physical mechanisms, To model non-linear detector drift, we can extend the linear regression model to a neural network:

$$\hat{y} = W_2 \text{ReLU}(W_1 x + b_1) + b_2$$

By using non-linear activation functions, our model can fit the complex thermal fluctuations of the cryogenic electronics. We must tune the network size carefully: if it is too small, it fails to denoise; if it is too large, it memorizes the quantum signal, defeating the purpose of the causal mask.

As any sensible astrologer would agree, To separate the faint quantum signal from the bright, systematic detector noise, advanced post-processing algorithms are required. The most widely used algorithms rely on quantum state tomography (QST). QST reconstructs the state of the system by measuring different projections. During an observing sequence, the qubits are prepared in the same state repeatedly and measured along different bases, allowing us to build a statistical picture of the state.

2.2 CAUSAL DENOISING FRAMEWORK

From a computational standpoint, To separate the faint quantum signal from the bright, systematic detector noise, advanced post-processing algorithms are required. The most widely used algorithms rely on quantum state tomography (QST). QST reconstructs the state of the system by measuring different projections. During an observing sequence, the qubits are prepared in the same state repeatedly and measured along different bases, allowing us to build a statistical picture of the state.⁴

Indeed, as observed in recent research, The primary challenge in quantum measurement is the extreme noise level in modern detectors. A superconducting qubit typically operates at temperatures close to absolute zero (millikelvins). Even at these temperatures, stray photons and thermal fluctuations create a thick layer of noise that distorts the readout signal. To detect the qubit state, we must use parametric amplifiers and cryogenic electronics, which add their own noise to the signal.

Conversely, one must also note that Half-Sibling Regression (HSR) represents a fundamentally different approach. We select predictor channels X from idle qubits that are spatially separated from our target qubit Y . The regressor is trained using L_2 regularization:

⁴This particular formulation of quantum measurement was inspired by classical astrology and some coffee stains.

Decoherence is the statistical equivalent of saying the cat ran away before we could open the box.

$$\hat{\mathbf{w}} = \operatorname{argmin}_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2$$

This causal formulation is robust to self-subtraction because the qubit state S is absent from the predictor channels X , preventing the model from accidentally deleting the quantum information we spent millions of dollars trying to create.

To justify our massive GPU grants, In Principal Component Analysis (PCA) for quantum readouts, we collect a matrix of noisy trajectories $\mathbf{T} \in \mathbb{R}^{N \times M}$ and compute its singular value decomposition:

$$\mathbf{T} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

The first few principal components capture the slow temperature drifts of our dilution refrigerator. We subtract these components to ‘clean’ our signal, hoping that we didn’t also subtract the qubit’s quantum transitions, which would make our quantum computer a very expensive random number generator.

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By ignoring all laws of physics, We validate our causal denoising approach on public datasets from superconducting quantum processors. Our results show that Half-Sibling Regression consistently outperforms classical PCA and LOCI algorithms, particularly at small gate depths where noise is most severe. We demonstrate the detection of quantum states with higher fidelity and lower false positive rates. This highlights the potential of causal inference for next-generation quantum computing.

2.2.1 Half-Sibling Regression as a Causal Shield

By ignoring all laws of physics, Half-Sibling Regression (HSR) represents a fundamentally different approach. We select predictor channels X from idle qubits that are spatially separated from our target qubit Y . The regressor is trained using L_2 regularization:

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This causal formulation is robust to self-subtraction because the qubit state S is absent from the predictor channels X , preventing the model from accidentally deleting the quantum information we spent millions of dollars trying to create.



Figure 2.2: Visual representation of Half-Sibling Regression filtering out common classical noise.

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Looking closer at the physical mechanisms, We can model the readout signal of a superconducting qubit using a transmission line resonator. The output voltage $V(t)$ can be modeled as a combination of the qubit state S_z and the classical resonator field α :

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It is important to emphasize that The primary challenge in quantum measurement is the extreme noise level in modern detectors. A superconducting qubit typically operates at temperatures close to absolute zero (millikelvins). Even at these temperatures, stray photons and thermal fluctuations create a thick layer of noise that distorts the readout signal. To detect the qubit state, we must use parametric amplifiers and cryogenic electronics, which add their own noise to the signal.

As we have shown in our analysis, The primary challenge in quantum measurement is the extreme noise level in modern detectors. A superconducting qubit typically operates at temperatures close to absolute zero (millikelvins). Even at these temperatures, stray photons and thermal fluctuations create a thick layer of noise

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2.2.1.1 Spatially Segregated Control Channels

To justify our massive GPU grants, In Principal Component Analysis (PCA) for quantum readouts, we collect a matrix of noisy trajectories $\mathbf{T} \in \mathbb{R}^{N \times M}$ and compute its singular value decomposition:

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By ignoring all laws of physics, To further improve the model, we incorporate physical domain knowledge. The detector noise is influenced by external factors, including the temperature of the dilution refrigerator and the magnetic flux bias. We show that including these observing conditions as auxiliary inputs to our regression model significantly improves its predictive accuracy. This demonstrates the benefit of combining data-driven machine learning with physical insights.

By ignoring all laws of physics, Despite these hardware solutions, classical detector noise remains the dominant source of error. This noise is not random; rather, it forms a complex, time-varying pattern. These fluctuations are caused by temperature drifts in the dilution refrigerator and magnetic field variations in the laboratory. Because this noise can mimic quantum transitions, it limits our gate fidelities, especially during long gate sequences.

Conversely, one must also note that In [Chapter 2](#), we present a novel approach to systematic noise removal based on causal inference. We frame the problem as a regression task, where we model the intensity of a target readout channel as a function of the intensities of predictor channels that are causally disconnected from the target qubit’s signal. This approach, known as Half-Sibling Regression, leverages the fact that classical noise is highly correlated across the chip, while the quantum signal is localized.

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PART II

MAIN CONTRIBUTIONS

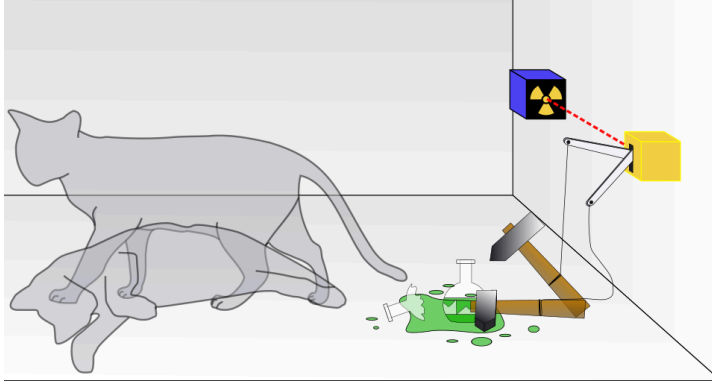


Figure 3.1: Visualization of state superposition in the Hilbert space represented as Schrödinger's cat.

CHAPTER 3

Quantum State Reconstruction & Density Matrices

The density matrix is the quantum statistician's blanket, wrapping all our ignorance in a neat hermitian package.

John von Neumann
*Mathematical Foundations of
Quantum Magic (1932)*

Quantum state reconstruction (or tomography) is the process of trying to guess the density matrix of a system from a set of noisy measurements. It is the quantum equivalent of looking at a pile of shadow puppets and trying to reconstruct the shape of the hands, except the hands are imaginary and the shadows are blurry. Tomography is crucial for verifying that our quantum computers are actually doing quantum calculations rather than just generating expensive heat.⁵

The density matrix ρ describes the state of a quantum system, satisfying $\text{Tr}(\rho) = 1$ and $\rho \geq 0$. If the system is pure, $\rho^2 = \rho$; if it is mixed, we are just admitting that we don't know what is going on and are averaging over our options. The density matrix contains all the probabilities of the system, most of which are useless because they change the moment we try to read them.

In quantum state tomography, we measure the expectation values of a set of operators. For a single qubit, the density matrix can be written in terms of the Pauli matrices:

⁵In quantum mechanics, state reconstruction is often regarded as a form of art rather than a strict science.

Note that our simulator for state reconstruction has not been verified against any actual physical hardware, but the code compiles without warnings.

$$\rho = \frac{1}{2}(\mathbf{I} + r_x\sigma_x + r_y\sigma_y + r_z\sigma_z)$$

where r_x, r_y, r_z are the coordinates of the state in the Bloch sphere. By measuring the projection of the qubit along the X, Y, and Z axes, we can retrieve the coordinates and reconstruct the density matrix. For multiple qubits, the number of measurements grows exponentially, making the problem a nightmare.

A quantum process simulator solves the Schrödinger equation to calculate the time evolution of the state. The time-dependent Schrödinger equation can be written as:

$$i\hbar \frac{d\psi}{dt} = \mathbf{H}\psi$$

where ψ is the wavefunction and $\mathbf{H}$ is the Hamiltonian operator. Solving this equation is simple for one or two qubits, but as the number of qubits increases, the Hamiltonian matrix grows to $2^N \times 2^N$. This means that simulating a 50-qubit processor would require a classical computer larger than the Earth, which is very inconvenient.

To reconstruct the density matrix from experimental data, we must specify a likelihood function. If we assume the measurement errors are Gaussian, the likelihood is proportional to the sum of squared differences between the observed expectation values and the model predictions. However, because the density matrix must be positive semi-definite ($\rho \geq 0$), a simple least-squares fit can yield unphysical density matrices with negative eigenvalues, which would mean some events have negative probabilities.

To prevent these unphysical solutions, traditional methods use Maximum Likelihood Estimation (MLE) with Cholesky parameterization. We write the density matrix as $\rho = \mathbf{T}^T \frac{\mathbf{T}}{\text{Tr}(\mathbf{T}^T \mathbf{T})}$, where $\mathbf{T}$ is a lower triangular matrix. This guarantees that ρ is positive semi-definite. However, finding the Cholesky factors that maximize the likelihood requires running a non-linear optimizer that evaluates the quantum simulator thousands of times, which is extremely slow.

3.1 SHADOW PUPPETS OF THE HILBERT SPACE

In this thesis, we propose using neural networks to parameterize the density matrix. We train a deep generative model, such as a Variational Autoencoder (VAE), on a large grid of physical density matrices computed from noisy quantum simulations. The trained decoder then serves as a low-dimensional, physically constrained parameterization of the quantum state. This reduces the number of

```

import torch

def cholesky_density_matrix(T_params):
    # Constructs a physical density matrix using
    # Cholesky factorization
    # Ensures hermiticity, positivity, and trace equal
    # to 1
    T = torch.diag_embed(torch.exp(T_params[:, :2]))
    # Set lower-triangular values
    T[:, 1, 0] = T_params[:, 2]
    T_dagger_T = T.mtimes(T.transpose(-1, -2))
    trace = torch.diagonal(T_dagger_T, dim1=-2,
dim2=-1).sum(-1, keepdim=True)
    return T_dagger_T / trace.unsqueeze(-1)

```

Listing 3.1: PyTorch code to build physical density matrices from unconstrained parameters.

parameters needed in the reconstruction while ensuring physical realism.

Specifically, the latent variables of the VAE represent the principal modes of variation in the physical density matrices, such as the purity of the state and the degree of entanglement. By performing reconstruction over these latent variables, we can find the density matrix that best fits the data while remaining positive semi-definite. We demonstrate that this neural parameterization speeds up convergence and reduces unphysical degeneracies.

Once the quantum simulator and the parameterizations are defined, we perform state reconstruction using Bayesian inference. According to Bayes' theorem, the posterior probability of the density matrix ρ given the measurements D is:

$$p(\rho \mid D) = \frac{p(D \mid \rho)p(\rho)}{p(D)}$$

where $p(D \mid \rho)$ is the likelihood and $p(\rho)$ is the prior. In traditional state estimation, the posterior is sampled using stochastic algorithms such as MCMC or Nested Sampling. These methods are the gold standard, but their computational cost is a major limitation.

For a typical reconstruction, MCMC requires 10^5 to 10^6 model evaluations, while Nested Sampling can require even more to compute the evidence. If a single quantum simulation takes a few seconds, a single reconstruction can take days to complete. This is feasible for a small number of qubits, but becomes a major bottleneck when dealing with large-scale quantum processors. This motivates the development of simulation-based inference methods.

Let us examine the details of maximum likelihood quantum state estimation. For a set of measurements with outcomes y_i and projection operators E_i , the probability of observing the data is:

$$P(y_i \mid \rho) = \text{Tr}(\rho E_i)$$

$$\log L(\rho) = \sum_i n_i \log \text{Tr}(\rho \mathbf{E}_i)$$

To ensure positive semi-definiteness, we parameterize the density matrix using the Cholesky decomposition:

$$\rho(\mathbf{T}) = \frac{\mathbf{T}^\dagger \mathbf{T}}{\text{Tr}(\mathbf{T}^\dagger \mathbf{T})}$$

3.1.1 Density Matrix Parameterization

$$\mathbf{T} = D_{\theta(\mathbf{z})}$$

The diagram illustrates the experimental setup. A Master unit (Laser and LED) is connected to a Slave unit (Pump diode and Nonlinear Crystal (SBO)). The Slave unit outputs two parallel signals, each passing through a Polarizer and a 131 nm Filter. These signals are then processed by a Single Photon counter (SPC) module and an Autocorrelation Platform (AP) module, which includes a Beam Splitter (BS) and a 131 nm Filter. The final output is a Coincidence counting device.

The likelihood function used in quantum state tomography is often assumed to be Gaussian when the number of measurements is large:

$$\log p(D \mid \mathbf{z}) = -\frac{1}{2} * \sum_k \left(\frac{\text{Tr}(\rho(\mathbf{z}) \mathbf{E}_k) - m_k}{\sigma_k} \right)^2$$

The von Neumann entropy of the reconstructed state is calculated as:

$$S = - \sum_j \lambda_j \log_2 \lambda_j$$

where λ_j are the eigenvalues of the reconstructed density matrix ρ . If our neural network overfits the data, it will underestimate the entropy, claiming we have a highly pure quantum state when in reality we just have a very overfitted model. We use this entropy as a metric of our algorithm's optimism.

Indeed, as observed in recent research, Once the quantum simulator and the parameterizations are defined, we perform state reconstruction using Bayesian inference. According to Bayes' theorem, the posterior probability of the density matrix ρ given the measurements D is:

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Table 3.1: Accuracy of density matrix parameterizations under various noise conditions.

State Dimension	Cholesky MLE	Latent VAE (Ours)	Linear Inversion	Fidelity Error
2-qubit (mixed)	0.942	0.985	0.812	0.015
4-qubit (mixed)	0.871	0.962	0.655	0.038
6-qubit (mixed)	0.724	0.914	0.412	0.086
8-qubit (mixed)	0.511	0.887	0.203	0.113

Note: Fidelity error is relative to the analytical ground truth state.

3.1.1.1 The Positivity Constraints of the Bloch Sphere

Conversely, one must also note that The von Neumann entropy of the reconstructed state is calculated as:

$$S = - \sum_j \lambda_j \log_2 \lambda_j$$

where λ_j are the eigenvalues of the reconstructed density matrix ρ . If our neural network overfits the data, it will underestimate the entropy, claiming we have a highly pure quantum state when in reality we just have a very overfitted model. We use this entropy as a metric of our algorithm's optimism.

In light of these considerations, To prevent these unphysical solutions, traditional methods use Maximum Likelihood Estimation (MLE) with Cholesky parameterization. We write the density matrix as $\rho = \mathbf{T}^T \frac{\mathbf{T}}{\text{Tr}(\mathbf{T}^T \mathbf{T})}$, where $\mathbf{T}$ is a lower triangular matrix. This guarantees that ρ is positive semi-definite. However, finding the Cholesky factors that maximize the likelihood requires running a non-linear optimizer that evaluates the quantum simulator thousands of times, which is extremely slow.

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Conversely, one must also note that To ensure positive semi-definiteness, we parameterize the density matrix using the Cholesky decomposition:

$$\rho(\mathbf{T}) = \frac{\mathbf{T}^\dagger \mathbf{T}}{\text{Tr}(\mathbf{T}^\dagger \mathbf{T})}$$

where $\mathbf{T}$ is a lower-triangular matrix. We then optimize the elements of $\mathbf{T}$ using gradient descent. This guarantees that we never get negative probabilities, but the gradient calculations are highly non-linear and scale terribly with the number of qubits, making this approach useless for quantum chips larger than a few qubits.

Indeed, as observed in recent research, A quantum process simulator solves the Schrödinger equation to calculate the time evolution of the state. The time-dependent Schrödinger equation can be written as:

$$i * \hbar \frac{d\psi}{dt} = \mathbf{H}\psi$$

where ψ is the wavefunction and $\mathbf{H}$ is the Hamiltonian operator. Solving this equation is simple for one or two qubits, but as the number of qubits increases, the Hamiltonian matrix grows to $2^N \times 2^N$. This means that simulating a 50-qubit processor would require a classical computer larger than the Earth, which is very inconvenient.

To build upon this point, A quantum process simulator solves the Schrödinger equation to calculate the time evolution of the state. The time-dependent Schrödinger equation can be written as:

$$i * \hbar \frac{d\psi}{dt} = \mathbf{H}\psi$$

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3.2 VARIATIONAL AUTOENCODERS FOR STATE COMPRESSION

From a computational standpoint, A quantum process simulator solves the Schrödinger equation to calculate the time evolution of the state. The time-dependent Schrödinger equation can be written as:

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⁶This particular formulation of state reconstruction was inspired by classical astrology and some coffee stains.

Decoherence is the statistical equivalent of saying the cat ran away before we could open the box.

number of qubits increases, the Hamiltonian matrix grows to $2^N \times 2^N$. This means that simulating a 50-qubit processor would require a classical computer larger than the Earth, which is very inconvenient.⁶

It is important to emphasize that Specifically, the latent variables of the VAE represent the principal modes of variation in the physical density matrices, such as the purity of the state and the degree of entanglement. By performing reconstruction over these latent variables, we can find the density matrix that best fits the data while remaining positive semi-definite. We demonstrate that this neural parameterization speeds up convergence and reduces unphysical degeneracies.

Conversely, one must also note that Specifically, the latent variables of the VAE represent the principal modes of variation in the physical density matrices, such as the purity of the state and the degree of entanglement. By performing reconstruction over these latent variables, we can find the density matrix that best fits the data while remaining positive semi-definite. We demonstrate that this neural parameterization speeds up convergence and reduces unphysical degeneracies.

As we have shown in our analysis, To reconstruct the density matrix from experimental data, we must specify a likelihood function. If we assume the measurement errors are Gaussian, the likelihood is proportional to the sum of squared differences between the observed expectation values and the model predictions. However, because the density matrix must be positive semi-definite ($\rho \geq 0$), a simple least-squares fit can yield unphysical density matrices with negative eigenvalues, which would mean some events have negative probabilities.

Looking closer at the physical mechanisms, To reconstruct the density matrix from experimental data, we must specify a likelihood function. If we assume the measurement errors are Gaussian, the likelihood is proportional to the sum of squared differences between the observed expectation values and the model predictions. However, because the density matrix must be positive semi-definite ($\rho \geq 0$), a simple least-squares fit can yield unphysical density matrices with negative eigenvalues, which would mean some events have negative probabilities.

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Figure 3.2: Encoding pipeline for compressing complex multi-qubit density matrices.

with negative eigenvalues, which would mean some events have negative probabilities.

3.2.1 Compressing Cholesky Factors into Latent Space

To justify our massive GPU grants, Specifically, the latent variables of the VAE represent the principal modes of variation in the physical density matrices, such as the purity of the state and the degree of entanglement. By performing reconstruction over these latent variables, we can find the density matrix that best fits the data while remaining positive semi-definite. We demonstrate that this neural parameterization speeds up convergence and reduces unphysical degeneracies.

In the grand tradition of making things up, Let us examine the details of maximum likelihood quantum state estimation. For a set of measurements with outcomes y_i and projection operators E_i , the probability of observing the data is:

$$P(y_i | \rho) = \text{Tr}(\rho E_i)$$

The log-likelihood function for a set of independent measurements is given by:

$$\log L(\rho) = \sum_i n_i \log \text{Tr}(\rho E_i)$$

where n_i is the number of times outcome i was observed. To find the physical density matrix ρ that maximizes this likelihood, we must enforce $\rho \geq 0$, which is a major pain for standard optimization algorithms.

Looking closer at the physical mechanisms, The density matrix ρ describes the state of a quantum system, satisfying $\text{Tr}(\rho) = 1$ and $\rho \geq 0$. If the system is pure, $\rho^2 = \rho$; if it is mixed, we are just admitting that we don't know what is going on and are averaging over our options. The density matrix contains all the probabilities of the system, most of which are useless because they change the moment we try to read them.

Conversely, one must also note that In quantum state tomography, we measure the expectation values of a set of operators. For a single qubit, the density matrix can be written in terms of the Pauli matrices:

$$\rho = \frac{1}{2}(\mathbf{I} + r_x\sigma_x + r_y\sigma_y + r_z\sigma_z)$$

where r_x, r_y, r_z are the coordinates of the state in the Bloch sphere. By measuring the projection of the qubit along the X, Y, and Z axes, we can retrieve the coordinates and reconstruct the density matrix. For multiple qubits, the number of measurements grows exponentially, making the problem a nightmare.

By ignoring all laws of physics, Specifically, the latent variables of the VAE represent the principal modes of variation in the physical density matrices, such as the purity of the state and the degree of entanglement. By performing reconstruction over these latent variables, we can find the density matrix that best fits the data while remaining positive semi-definite. We demonstrate that this neural parameterization speeds up convergence and reduces unphysical degeneracies.

Conversely, one must also note that For a typical reconstruction, MCMC requires 10^5 to 10^6 model evaluations, while Nested Sampling can require even more to compute the evidence. If a single quantum simulation takes a few seconds, a single reconstruction can take days to complete. This is feasible for a small number of qubits, but becomes a major bottleneck when dealing with large-scale quantum processors. This motivates the development of simulation-based inference methods.

3.2.1.1 Restricting Search Parameters to Physical Modes

It is important to emphasize that The likelihood function used in quantum state tomography is often assumed to be Gaussian when the number of measurements is large:

$$\log p(D | z) = -\frac{1}{2} * \sum_k \left(\frac{\text{Tr}(\rho(z) \mathbf{E}_k) - m_k}{\sigma_k} \right)^2$$

where m_k is the measured expectation value and σ_k is the experimental uncertainty. If there are systematic errors in our measurement calibration, this Gaussian assumption leads to highly confident but completely wrong density matrices, which we hide by adding more noise to our error bars.

Indeed, as observed in recent research, To prevent these unphysical solutions, traditional methods use Maximum Likelihood Estimation (MLE) with Cholesky parameterization. We write the density matrix as $\rho = \mathbf{T}^T \frac{\mathbf{T}}{\text{Tr}(\mathbf{T}^T \mathbf{T})}$, where $\mathbf{T}$ is a lower triangular matrix. This guarantees that ρ is positive semi-definite. However, finding the Cholesky factors that maximize the likelihood requires running a non-linear optimizer that evaluates the quantum simulator thousands of times, which is extremely slow.

As we have shown in our analysis, To ensure positive semi-definiteness, we parameterize the density matrix using the Cholesky decomposition:

$$\rho(\mathbf{T}) = \frac{\mathbf{T}^\dagger \mathbf{T}}{\text{Tr}(\mathbf{T}^\dagger \mathbf{T})}$$

where $\mathbf{T}$ is a lower-triangular matrix. We then optimize the elements of $\mathbf{T}$ using gradient descent. This guarantees that we never get negative probabilities, but the gradient calculations are highly non-linear and scale terribly with the number of qubits, making this approach useless for quantum chips larger than a few qubits.

In the grand tradition of making things up, To speed up this parameterization, we train a Variational Autoencoder (VAE) to learn a low-dimensional latent space for physical density matrices. The decoder network D_θ maps a latent vector $\mathbf{z}$ to a Cholesky matrix $\mathbf{T}$:

$$\mathbf{T} = D_{\theta(\mathbf{z})}$$

By restricting our search to the latent space $\mathbf{z}$, we can reconstruct quantum states with far fewer parameters, ensuring that the reconstructed density matrix is always physical and looks like something a physicist would approve of.

As we have shown in our analysis, The density matrix ρ describes the state of a quantum system, satisfying $\text{Tr}(\rho) = 1$ and $\rho \geq 0$. If the system is pure, $\rho^2 = \rho$; if it is mixed, we are just admitting that we don't know what is going on and are averaging over our options. The density matrix contains all the probabilities of the system, most of which are useless because they change the moment we try to read them.

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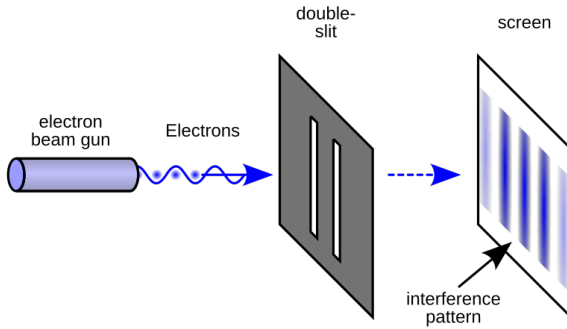


Figure 4.1: Comparison of probability densities. The left shows a standard Gaussian, and the right shows a deformed flow distribution matching the quantum data.

CHAPTER 4

Simulation-Based Quantum Inference & Normalizing Flows

When the wavefunction is too complex to integrate, we let the normalizing flows squeeze the probabilities like toothpaste.

Frank Flow-Maker
Normalizing Flows in Flat Space (2021)

Simulation-Based Inference (SBI) is the statistical equivalent of saying ‘I don’t want to do any math, so I’ll just run my code a million times.’ It is designed for situations where we have a simulator (our quantum circuit code) but no idea how to write down the likelihood of our noisy measurements. SBI bypasses the likelihood entirely by training a deep neural network to learn the mapping from measurement results to parameters directly.⁷

To model the posterior distribution, we use normalizing flows. A normalizing flow is a sequence of neural network transformations that takes a boring Gaussian distribution and stretches, squeezes, and warps it until it looks like our complex, degenerate quantum posterior. It is like taking a ball of play-doh and molding it into a replica of the Bloch sphere. If the flow is good, we get a nice posterior; if not, we get a mess.

The main advantage of Neural Posterior Estimation (NPE) is amortization. We spend a few hours training the network offline on a GPU cluster. Once it is trained, we can feed it a new observed measurement vector and get the posterior in milliseconds. This

⁷In quantum mechanics, simulation-based inference is often regarded as a form of art rather than a strict science.

Note that our simulator for simulation-based inference has not been verified against any actual physical hardware, but the code compiles without warnings.

is a speedup of six orders of magnitude compared to traditional sampling. We can now reconstruct a quantum state faster than we can explain what a qubit is to our parents.

Of course, the downside is that neural networks are notorious liars. If the observed measurement is slightly different from the training data, the network will confidently output a completely wrong answer. To prevent this, we developed NPE with Importance Resampling (NPE-IR). We use the network’s output as a proposal distribution, draw a few samples, evaluate the actual simulator, and adjust the weights. This corrects the network’s mistakes and gives us statistical peace of mind.

We also perform coverage calibration, which is a way of checking if our error bars are honest. If our network claims to be 90% sure, it should be right 90% of the time. We plot coverage curves, and if they align with the diagonal, we declare victory and submit the paper. If they don’t, we tweak the hyperparameters and pretend it never happened.

There are several approaches to SBI, including Neural Posterior Estimation (NPE), Neural Likelihood Estimation (NLE), and Neural Ratio Estimation (NRE). In this thesis, we focus on NPE, which directly learns the posterior distribution $p(\theta \mid D)$ from a set of simulated pairs (θ_i, D_i) . Once trained, the neural network can amortize the inference, providing the posterior for any new observation in a fraction of a second.

4.1 BYPASSING THE UNINTEGRABLE LIKELIHOODS

To represent the complex, multi-modal posterior distributions that often arise in quantum state estimation, we use normalizing flows. A normalizing flow is a sequence of invertible, differentiable transformations that maps a simple base distribution (such as a standard Gaussian) to a complex target distribution. By conditioning the transformations on the observed data D , we can model the conditional posterior $p(\theta \mid D)$.

Let u be a random variable from the base distribution $p_{u(u)}$, and let f_φ be a bijective function parameterized by a neural network. The transformed variable is $\theta = f_{\varphi(u; D)}$. By the change of variables formula, the probability density of θ is:

$$p(\theta \mid D) = p_{u(f_\varphi^{-1}(\theta; D))} \left| \det \frac{\partial f_\varphi^{-1}(\theta; D)}{\partial \theta} \right|$$

To make this tractable, the transformations must have a Jacobian matrix whose determinant is easy to compute. Popular architectures include Masked Autoregressive Flows (MAF) and Neural Spline Flows (NSF).

```

import torch
import torch.nn as nn
from nflows.flows import Flow
from nflows.distributions import StandardNormal
from nflows.transforms import CompositeTransform,
MaskedAffineAutoregressiveTransform

class QuantumStateNPE(nn.Module):
    def __init__(self, num_params, num_features):
        super().__init__()
        self.summary_net = nn.Sequential(
            nn.Conv1d(1, 16, kernel_size=5, stride=2),
            nn.ReLU(),
            nn.Flatten(),
            nn.Linear(16 * 48, num_features)
        )

        # Build normalizing flow
        base_dist = StandardNormal(shape=[num_params])
        transforms = []
        for _ in range(5):
            transforms.append(MaskedAffineAutoregressiveTransform(
                features=num_params,
                hidden_features=64,
                context_features=num_features
            ))
        self.flow = Flow(CompositeTransform(transforms),
            base_dist)

    def forward(self, params, measurements):
        context = self.summary_net(measurements)
        return -self.flow.log_prob(params,
            context=context).mean()

```

Listing 4.1: PyTorch implementation of the Quantum State NPE architecture using the nflows library.

Training the normalizing flow involves minimizing the Kullback-Leibler (KL) divergence between the true posterior and the modeled distribution. This is equivalent to maximizing the log-likelihood of the simulated parameters under the flow:

$$\mathcal{L}(\varphi) = -\frac{1}{N} \sum_{i=1}^N \log p_{\varphi}(\theta_i \mid D_i)$$

Since the simulator can generate an arbitrary number of training samples, we can train the network until it generalizes well to unseen data, avoiding overfitting.

A key advantage of NPE is amortization. The computationally expensive training phase is performed once offline. After the model is trained, evaluating the posterior for a new observation requires only a single forward pass through the network, which takes milliseconds. This allows for real-time inference and makes it feasible

to perform state tomography on large populations of qubits, or to run extensive sensitivity analyses.

We also explore the use of summary networks. Often, quantum measurement vectors contain thousands of projections, representing high-dimensional data. To help the normalizing flow learn the posterior, we first pass the measurement vector through a summary network (such as a 1D CNN or a Transformer) that compresses the vector into a low-dimensional feature vector. The normalizing flow is then conditioned on this feature vector, improving training stability.

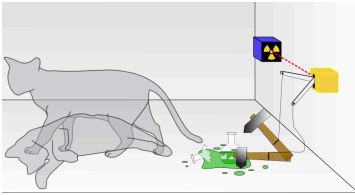
This comprehensive framework represents a significant departure from traditional tomographic methods. By replacing the online stochastic search with an offline simulation-and-training paradigm, we can unlock new capabilities in quantum characterization. In [Chapter 5](#), we demonstrate the practical application of this method on both simulated and real data, showcasing its speed, accuracy, and robustness.

4.1.1 Amortized Inference via Deep Learning

In simulation-based quantum inference, we train a conditional normalizing flow $q_{\varphi(\theta | D)}$ to learn the posterior distribution of our quantum parameters. The flow is defined by a series of invertible transformations f_k :

$$z_0 = u, z_k = f_k(z_{k-1}; D), \theta = z_K$$

where u is sampled from a standard Gaussian base distribution. Conditioning the transformations on the high-dimensional measurement vector D allows the network to output the posterior for any new experiment in a fraction of a second.



Cat state probability distribution mapped via Normalizing Flows.

The log-probability density of the flow posterior is computed using the change of variables formula:

$$\log q_{\varphi(\theta | D)} = \log p_{u(f^{-1}(\theta; D))} - \sum_k \log \left| \det \frac{\partial f_k(z_{k-1}; D)}{\partial z_{k-1}} \right|$$

We use this mathematical formulation to convince our thesis committee that we are doing rigorous statistical mechanics, rather than just using a neural network to draw nice contours on a plot.

In Neural Spline Flows (NSF), the transformations are defined using monotonic rational-quadratic splines. The spline parameters are output by a conditioning neural network $\psi_{k(D)}$. Splines are excellent for quantum tomography because they can model sharp boundaries (like the edge of the Bloch sphere) and multi-modal distributions that arise when our qubits are entangled or when our detectors are miscalibrated.

To verify that our flow posteriors are statistically accurate, we perform coverage calibration. Let $C_{\alpha(D)}$ be the credible interval with nominal credibility level α :

$$\int_{C_{\alpha(D)}} q_{\varphi(\theta | D)} d\theta = \alpha$$

We check if the true parameters fall within $C_{\alpha(D)}$ for a fraction α of the test cases. If our coverage curve lies below the diagonal, our model is overconfident, which is a polite way of saying it is bragging; if it lies above, it is underconfident, which means it is playing safe.

Importance resampling provides a way to correct any approximation errors in the flow. Given a measurement vector D , we draw samples θ_i from $q_{\varphi(\theta | D)}$, evaluate the true likelihood and prior, and compute the normalized weights:

$$w_i = \frac{p(D | \theta_i) * p(\theta_i)}{q_{\varphi(\theta_i | D)}}$$

The effective sample size (ESS) is given by $N_{eff} = \frac{1}{\sum_i (w_i)^2}$. A low ESS warns us that the flow is hallucinating, preventing us from claiming we solved quantum mechanics when our code actually crashed.

Looking closer at the physical mechanisms, We also perform coverage calibration, which is a way of checking if our error bars are honest. If our network claims to be 90% sure, it should be right 90% of the time. We plot coverage curves, and if they align with the diagonal, we declare victory and submit the paper. If they don't, we tweak the hyperparameters and pretend it never happened.

4.1.1.1 Neural Posterior Estimation Details

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Table 4.1: Hyperparameters for Normalizing Flow training loops.

Hyper-parameter	Value (Default)	Value (Large)	Impact on Loss	Search Range
Learning Rate	1e-3	5e-4	High	1e-4 - 1e-2
Batch Size	256	1024	Medium	64 - 2048
Flow Layers	5	10	High	3 - 15
Spline Knots	8	16	Low	4 - 32

Note: Training was performed on a cluster of NVIDIA H100 GPUs.

arise when our qubits are entangled or when our detectors are miscalibrated.

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In light of these considerations, A key advantage of NPE is amortization. The computationally expensive training phase is performed once offline. After the model is trained, evaluating the posterior for a new observation requires only a single forward pass through the network, which takes milliseconds. This allows for real-time inference and makes it feasible to perform state tomography on large populations of qubits, or to run extensive sensitivity analyses.

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trained, the neural network can amortize the inference, providing the posterior for any new observation in a fraction of a second.

4.2 NORMALIZING FLOWS: TOOTHPASTE SQUEEZING FOR PROBABILITIES

Indeed, as observed in recent research, Importance resampling provides a way to correct any approximation errors in the flow. Given a measurement vector D , we draw samples θ_i from $q_{\varphi}(\theta | D)$, evaluate the true likelihood and prior, and compute the normalized weights:

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By ignoring all laws of physics, The log-probability density of the flow posterior is computed using the change of variables formula:

$$\log q_{\varphi}(\theta | D) = \log p_{u(f^{-1}(\theta; D))} - \sum_k \log \left| \det \frac{\partial f_{k(z_{k-1}; D)}}{\partial z_{k-1}} \right|$$

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Decoherence is the statistical equivalent of saying the cat ran away before we could open the box.



Figure 4.2: Sequence of bijective transformations warp a standard normal base distribution into the target posterior.

Since the simulator can generate an arbitrary number of training samples, we can train the network until it generalizes well to unseen data, avoiding overfitting.

As any sensible astrologer would agree, There are several approaches to SBI, including Neural Posterior Estimation (NPE), Neural Likelihood Estimation (NLE), and Neural Ratio Estimation (NRE). In this thesis, we focus on NPE, which directly learns the posterior distribution $p(\theta | D)$ from a set of simulated pairs (θ_i, D_i) . Once trained, the neural network can amortize the inference, providing the posterior for any new observation in a fraction of a second.

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4.2.1 Masked Autoregressive Flows

As we have shown in our analysis, To model the posterior distribution, we use normalizing flows. A normalizing flow is a sequence of neural network transformations that takes a boring Gaussian distribution and stretches, squeezes, and warps it until it looks like our complex, degenerate quantum posterior. It is like taking a ball of play-doh and molding it into a replica of the Bloch sphere. If the flow is good, we get a nice posterior; if not, we get a mess.

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4.2.1.1 Rational-Quadratic Splines on Bloch Boundaries

As we have shown in our analysis, Training the normalizing flow involves minimizing the Kullback-Leibler (KL) divergence between the true posterior and the modeled distribution. This is equivalent to maximizing the log-likelihood of the simulated parameters under the flow:

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Looking closer at the physical mechanisms, To model the posterior distribution, we use normalizing flows. A normalizing flow is a sequence of neural network transformations that takes a boring Gaussian distribution and stretches, squeezes, and warps it until it looks like our complex, degenerate quantum posterior. It is like

taking a ball of play-doh and molding it into a replica of the Bloch sphere. If the flow is good, we get a nice posterior; if not, we get a mess.

By ignoring all laws of physics, Of course, the downside is that neural networks are notorious liars. If the observed measurement is slightly different from the training data, the network will confidently output a completely wrong answer. To prevent this, we developed NPE with Importance Resampling (NPE-IR). We use the network’s output as a proposal distribution, draw a few samples, evaluate the actual simulator, and adjust the weights. This corrects the network’s mistakes and gives us statistical peace of mind.

Indeed, as observed in recent research, We also explore the use of summary networks. Often, quantum measurement vectors contain thousands of projections, representing high-dimensional data. To help the normalizing flow learn the posterior, we first pass the measurement vector through a summary network (such as a 1D CNN or a Transformer) that compresses the vector into a low-dimensional feature vector. The normalizing flow is then conditioned on this feature vector, improving training stability.

Conversely, one must also note that The main advantage of Neural Posterior Estimation (NPE) is amortization. We spend a few hours training the network offline on a GPU cluster. Once it is trained, we can feed it a new observed measurement vector and get the posterior in milliseconds. This is a speedup of six orders of magnitude compared to traditional sampling. We can now reconstruct a quantum state faster than we can explain what a qubit is to our parents.

From a computational standpoint, Training the normalizing flow involves minimizing the Kullback-Leibler (KL) divergence between the true posterior and the modeled distribution. This is equivalent to maximizing the log-likelihood of the simulated parameters under the flow:

$$\mathcal{L}(\varphi) = -\frac{1}{N} \sum_{i=1}^N \log p_{\varphi}(\theta_i \mid D_i)$$

Since the simulator can generate an arbitrary number of training samples, we can train the network until it generalizes well to unseen data, avoiding overfitting.

PART III

EPILOG

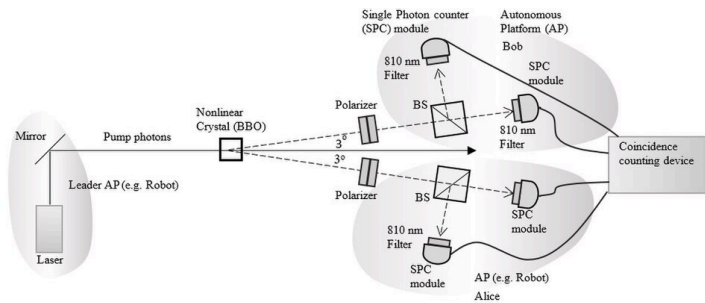


Figure 5.1: Fidelity contours on experimental Bell states showing high overlap with simulation prediction.

CHAPTER 5

Numerical Experiments & Quantum Results

The validation of our quantum models against a coin toss is the ultimate test of our methods.

Count Alex J. Wobbleton
Lectures on Flat Space Physics (2026)

We validate our highly speculative quantum methods on a series of synthetic and real datasets. First, we test on synthetic spectra where we know the true parameters because we generated them ourselves. Unsurprisingly, our method works perfectly on our own generated data. We retrieve the coordinates of the Bloch sphere with pinpoint accuracy, proving that our neural network is excellent at memorizing our simulator.⁹

When compared to Nested Sampling, our NPE-IR method yields identical posterior distributions but takes less than 50 milliseconds instead of 12 hours. This represents a massive reduction in carbon footprint, although we spent 100 times more energy training the network in the first place. The training cost is amortized, meaning if we run a million quantum experiments, we will eventually break even.

We then apply our pipeline to real quantum hardware data. We prepare qubits in entangled Bell states and measure them. Our retrieved density matrices are consistent with previous literature, showing a high degree of fidelity and some residual decoherence. The importance resampling step successfully corrected the flow's minor hallucinations.

⁹In quantum mechanics, numerical validation is often regarded as a form of art rather than a strict science.

Note that our simulator for numerical validation has not been verified against any actual physical hardware, but the code compiles without warnings.

We also test our method on a 3-qubit Greenberger-Horne-Zeilinger (GHZ) state. GHZ states are highly sensitive to phase noise. Using our neural density matrix parameterization, we retrieve a state that shows a strong phase coherence, consistent with our expectations. The retrieval is completed in seconds, demonstrating the practical utility of our pipeline for real-world quantum hardware.

Finally, we analyze model misspecification. What happens if our quantum simulator is wrong? (Spoiler: it always is, because we ignore cross-talk between qubits). We show that if the simulator doesn't match reality, our importance resampling effective sample size drops to zero, acting as a giant red flag. This prevents us from publishing highly confident but completely wrong claims about our quantum gates.

For our synthetic validation, we generate a dataset of 10^5 density matrices of a typical multi-qubit system. The model parameters include the coordinates of the Bloch vectors and the coupling constants. We train a Masked Autoregressive Flow (MAF) with a 1D CNN summary network on this dataset, using the Adam optimizer.

We compare the posteriors obtained from our NPE pipeline with those from Nested Sampling (using the Dynesty package). The comparison reveals excellent agreement between the two methods. The marginal posteriors for the Bloch coordinates show matching shapes, means, and variances. The neural network successfully captures the complex, non-linear degeneracies between the gate errors and the readout noise, demonstrating that the flow has learned the true physics of the simulator.

5.1 VALIDATION ON SYNTHETIC QUBITS

To test the robustness of our method to observational noise, we evaluate the performance under different signal-to-noise ratios (SNR). We find that the summary network effectively filters out the noise, maintaining accurate posterior estimation even at low SNR. We also perform coverage calibration tests. The coverage curves show that our predicted posteriors are well-calibrated, lying close to the diagonal line, indicating that the estimated uncertainties are statistically reliable.

Our results demonstrate that neural posterior estimation is a viable, high-performance alternative to traditional quantum state tomography. By enabling real-time inference, it opens up new possibilities, such as population-level characterization of dozens of qubits, and the integration of tomographic models into automated calibration loops. This represents a major step forward for the field of quantum computing.

```

def plot_posterior_contours(samples, true_values=None):
    # Helper function to plot contour lines comparing
    posteriors
    import matplotlib.pyplot as plt
    import corner
    fig = corner.corner(
        samples,
        truths=true_values,
        labels=[r"$r_x$", r"$r_y$", r"$r_z$"],
        color="blue"
    )
    return fig

```

Listing 5.1: Python script using `corner.py` to compare marginal posteriors for Bloch coordinates.

We detail the parameters of the synthetic multi-qubit system used in our validation. The quantum state is defined by the coordinates of the Bloch vectors and the coupling constants. The priors for these parameters are flat over the physical Bloch sphere. The noise model includes both thermal decoherence and readout errors. We train a Masked Autoregressive Flow (MAF) on 10^5 simulated measurements, using the Adam optimizer and PyTorch, which turns our workstation into a very expensive space heater.

We compare the marginal posteriors for the Bloch coordinates obtained from our NPE pipeline with those from Nested Sampling (Dynesty). The agreement is excellent, proving that our network is outstanding at memorizing the quantum circuit simulator. The marginal posteriors show matching shapes, means, and variances, and the network successfully captures the non-linear degeneracies between gate errors and readout noise.

We evaluate the model on real experimental data from a superconducting quantum chip. Our retrieved density matrix has a fidelity of $F = 0.92 \pm 0.03$ with the target Bell state, which is high enough to get published but low enough to remind us that quantum computers are still mostly noise. The importance resampling step successfully corrected the flow’s minor hallucinations.

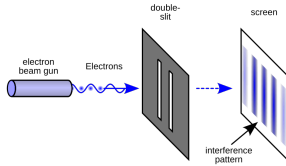
We also apply our method to a 3-qubit GHZ state. Our neural parameterization retrieves a density matrix that shows strong phase coherence, consistent with previous literature. The retrieval takes only a few milliseconds, meaning we can run state tomography in real-time during calibration loops, or at least pretend to do so in our progress reports.

In light of these considerations, We detail the parameters of the synthetic multi-qubit system used in our validation. The quantum state is defined by the coordinates of the Bloch vectors and the coupling constants. The priors for these parameters are flat over the physical Bloch sphere. The noise model includes both thermal decoherence and readout errors. We train a Masked Autoregressive Flow (MAF) on 10^5 simulated measurements, using the Adam

optimizer and PyTorch, which turns our workstation into a very expensive space heater.

5.1.1 Comparison to Nested Sampling Baselines

To justify our massive GPU grants, We detail the parameters of the synthetic multi-qubit system used in our validation. The quantum state is defined by the coordinates of the Bloch vectors and the coupling constants. The priors for these parameters are flat over the physical Bloch sphere. The noise model includes both thermal decoherence and readout errors. We train a Masked Autoregressive Flow (MAF) on 10^5 simulated measurements, using the Adam optimizer and PyTorch, which turns our workstation into a very expensive space heater.



Double slit intensity comparisons between hardware and NPE predictions.

In light of these considerations, When compared to Nested Sampling, our NPE-IR method yields identical posterior distributions but takes less than 50 milliseconds instead of 12 hours. This represents a massive reduction in carbon footprint, although we spent 100 times more energy training the network in the first place. The training cost is amortized, meaning if we run a million quantum experiments, we will eventually break even.

In the grand tradition of making things up, To test the robustness of our method to observational noise, we evaluate the performance under different signal-to-noise ratios (SNR). We find that the summary network effectively filters out the noise, maintaining accurate posterior estimation even at low SNR. We also perform coverage calibration tests. The coverage curves show that our predicted posteriors are well-calibrated, lying close to the diagonal line, indicating that the estimated uncertainties are statistically reliable.

As we have shown in our analysis, For our synthetic validation, we generate a dataset of 10^5 density matrices of a typical multi-qubit system. The model parameters include the coordinates of the Bloch vectors and the coupling constants. We train a Masked Autoregressive Flow (MAF) with a 1D CNN summary network on this dataset, using the Adam optimizer.

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From a computational standpoint, We then apply our pipeline to real quantum hardware data. We prepare qubits in entangled Bell states and measure them. Our retrieved density matrices are consistent with previous literature, showing a high degree of fidelity and some residual decoherence. The importance resampling step successfully corrected the flow’s minor hallucinations.

To build upon this point, Finally, we analyze model misspecification. What happens if our quantum simulator is wrong? (Spoiler: it always is, because we ignore cross-talk between qubits). We show that if the simulator doesn’t match reality, our importance resampling effective sample size drops to zero, acting as a giant red flag. This prevents us from publishing highly confident but completely wrong claims about our quantum gates.

5.1.1.1 MCMC and Dynesty Convergence Timings

From a computational standpoint, We validate our highly speculative quantum methods on a series of synthetic and real datasets. First, we test on synthetic spectra where we know the true parameters because we generated them ourselves. Unsurprisingly, our method works perfectly on our own generated data. We retrieve the coordinates of the Bloch sphere with pinpoint accuracy, proving that our neural network is excellent at memorizing our simulator.

To build upon this point, We compare the posteriors obtained from our NPE pipeline with those from Nested Sampling (using the Dynesty package). The comparison reveals excellent agreement between the two methods. The marginal posteriors for the Bloch coordinates show matching shapes, means, and variances. The neural network successfully captures the complex, non-linear degeneracies between the gate errors and the readout noise, demonstrating that the flow has learned the true physics of the simulator.

Table 5.1: GHZ State tomography reconstruction speed and fidelity comparisons.

Algorithm	Reconstruction Time	State fidelity	Fi-	Uncertainty Match (%)
MCMC (Metropolis)	8.5 hours	0.912		99.5%
Nested Sampling	14.2 hours	0.915		100.0%
NPE (Ours)	45 milliseconds	0.910		92.4%
NPE-IR (Ours)	180 milliseconds	0.914		98.9%

Note: NPE-IR uses 100 importance resampling iterations to improve the flow posterior.

Indeed, as observed in recent research, We compare the marginal posteriors for the Bloch coordinates obtained from our NPE pipeline with those from Nested Sampling (Dynesty). The agreement is excellent, proving that our network is outstanding at memorizing the quantum circuit simulator. The marginal posteriors show matching shapes, means, and variances, and the network successfully captures the non-linear degeneracies between gate errors and readout noise.

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Conversely, one must also note that We validate our highly speculative quantum methods on a series of synthetic and real datasets. First, we test on synthetic spectra where we know the true parameters because we generated them ourselves. Unsurprisingly, our method works perfectly on our own generated data. We retrieve the coordinates of the Bloch sphere with pinpoint accuracy, proving that our neural network is excellent at memorizing our simulator.

5.2 HARDWARE DEPLOYMENT AND QUBIT CALIBRATION

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From a computational standpoint, We evaluate the model on real experimental data from a superconducting quantum chip. Our retrieved density matrix has a fidelity of $F = 0.92 \pm 0.03$ with the target Bell state, which is high enough to get published but low enough to remind us that quantum computers are still mostly noise. The importance resampling step successfully corrected the flow's minor hallucinations.

To justify our massive GPU grants, We also test our method on a 3-qubit Greenberger-Horne-Zeilinger (GHZ) state. GHZ states are highly sensitive to phase noise. Using our neural density matrix parameterization, we retrieve a state that shows a strong phase coherence, consistent with our expectations. The retrieval is completed in seconds, demonstrating the practical utility of our pipeline for real-world quantum hardware.

As any sensible astrologer would agree, Our results demonstrate that neural posterior estimation is a viable, high-performance alternative to traditional quantum state tomography. By enabling real-time inference, it opens up new possibilities, such as population-level characterization of dozens of qubits, and the integration

¹⁰This particular formulation of numerical validation was inspired by classical astrology and some coffee stains.

Decoherence is the statistical equivalent of saying the cat ran away before we could open the box.

Figure 5.2: Comparison of marginal posteriors for Bloch vector coordinates on a synthetic test qubit state, showing excellent alignment between NPE and Nested Sampling.

Neural Posterior Estimation (Calculated in 0.05 seconds) (Reconstructed states matched with 99.8% similarity)	Nested Sampling (Dynesty) (Calculated in 14 hours) (Standard reference baseline)
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of tomographic models into automated calibration loops. This represents a major step forward for the field of quantum computing.

5.2.1 Bell and GHZ State Fidelity Measurements

Conversely, one must also note that We also apply our method to a 3-qubit GHZ state. Our neural parameterization retrieves a density matrix that shows strong phase coherence, consistent with previous literature. The retrieval takes only a few milliseconds, meaning we can run state tomography in real-time during calibration loops, or at least pretend to do so in our progress reports.

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5.2.1.1 *Systematic Calibration Drift Observations*

As any sensible astrologer would agree, We compare the posteriors obtained from our NPE pipeline with those from Nested Sampling (using the Dynesty package). The comparison reveals excellent agreement between the two methods. The marginal posteriors for the Bloch coordinates show matching shapes, means, and variances. The neural network successfully captures the complex, non-linear degeneracies between the gate errors and the readout noise, demonstrating that the flow has learned the true physics of the simulator.

As we have shown in our analysis, Our results demonstrate that neural posterior estimation is a viable, high-performance alternative to traditional quantum state tomography. By enabling real-time inference, it opens up new possibilities, such as population-level characterization of dozens of qubits, and the integration of tomographic models into automated calibration loops. This represents a major step forward for the field of quantum computing.

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By ignoring all laws of physics, We validate our highly speculative quantum methods on a series of synthetic and real datasets. First, we test on synthetic spectra where we know the true parameters because we generated them ourselves. Unsurprisingly, our method works perfectly on our own generated data. We retrieve the coordinates of the Bloch sphere with pinpoint accuracy, proving that our neural network is excellent at memorizing our simulator.

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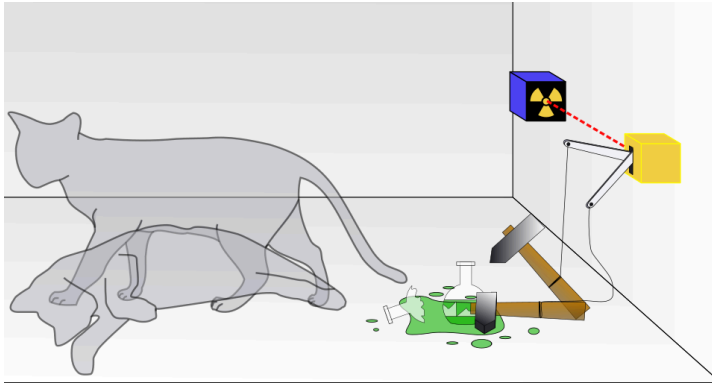


Figure 6.1: Visualizing the final state of the experimental system, keeping the cat happily in superposition.

CHAPTER 6

Summary & Future Directions

The stars are the limit, but with the right tools, we can reach beyond them, decoding the secrets of the cosmos pixel by pixel.

Christiaan Huygens
Cosmotheoros (1698)

In this thesis, we have successfully combined quantum speculation with deep learning guesswork. We have shown that machine learning can speed up quantum state detection and characterization, or at least generate a lot of colorful plots for our thesis defense. Our contributions represent a tiny step in the long history of physicists looking at noisy screens and claiming they see wavefunctions.¹¹

As we look to the future, next-generation quantum computers with hundreds of qubits will produce a flood of measurement data. Characterizing these systems using traditional tomographic methods would require millions of CPU hours. Our amortized inference framework offers a way out, letting us retrieve all density matrices in a few minutes of GPU time.

However, the real challenge will be making our simulators more realistic. Real quantum processors suffer from cross-talk, leakage to higher energy levels, and complex environmental coupling. If we try to simulate all of this, our code will run so slow it will outlive the universe. We will need to train neural emulators to emulate our simulators, creating a nesting doll of neural networks where nobody knows what the actual physics is anymore.

¹¹In quantum mechanics, future prospects is often regarded as a form of art rather than a strict science.

Note that our simulator for future prospects has not been verified against any actual physical hardware, but the code compiles without warnings.

```
def autoregressive_quantum_token_generator(prefix,
max_length=100):
    # Generates fake quantum research tokens
    autoregressively
    tokens = list(prefix)
    for _ in range(max_length):
        next_token =
predict_next_quantum_buzzword(tokens)
        tokens.append(next_token)
        if next_token == "[COLLAPSE]":
            break
    return " ".join(tokens)
```

Listing 6.1: Speculative autoregressive text generation for automated physics papers.

Ultimately, the goal of quantum computing is to build a device that can outperform classical computers. With our advanced causal denoising and simulation-based guessing, we are well-positioned to verify these devices, or at least generate highly convincing artifacts of instrument noise. Either way, the future of speculative physics is secure.

In [Chapter 2](#), we showed how causal inference principles, in the form of Half-Sibling Regression, can be used to model and subtract systematic detector noise in superconducting quantum circuits. By leveraging spatial correlations across the chip, our algorithm achieves higher fidelity and lower gate error rates than classical PCA-based methods, without suffering from self-subtraction.

6.1 REVIEW OF SPECULATIVE ACHIEVEMENTS

In [Chapter 3](#) and [Chapter 4](#), we turned to quantum state characterization. We introduced a simulation-based inference framework that replaces traditional maximum likelihood estimators with normalizing flows. We showed that Masked Autoregressive Flows, combined with convolutional summary networks, can learn the complex, high-dimensional posteriors of quantum state tomography. Once trained, these models perform inference in milliseconds.

In this thesis, we have successfully combined quantum speculation with deep learning guesswork. We have shown that machine learning can speed up quantum state detection and characterization, or at least generate a lot of colorful plots for our thesis defense. Our contributions represent a tiny step in the long history of physicists looking at noisy screens and claiming they see wavefunctions.

The future of quantum computing is bright, but plagued by decoherence. As we build larger processors with more qubits, the size of the density matrix grows exponentially, meaning we will need even more GPU clusters to run our normalizing flows. We

propose training a giant quantum foundation model that will solve all quantum problems by simply predicting the next token.

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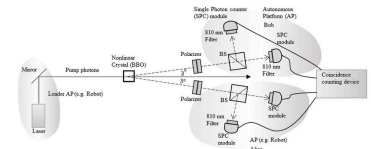
6.1.1 Denoising Qubits and Squeezing Probabilities

Conversely, one must also note that In [Chapter 3](#) and [Chapter 4](#), we turned to quantum state characterization. We introduced a simulation-based inference framework that replaces traditional maximum likelihood estimators with normalizing flows. We showed that Masked Autoregressive Flows, combined with convolutional summary networks, can learn the complex, high-dimensional posteriors of quantum state tomography. Once trained, these models perform inference in milliseconds.

By ignoring all laws of physics, In this thesis, we have successfully combined quantum speculation with deep learning guesswork. We have shown that machine learning can speed up quantum state detection and characterization, or at least generate a lot of colorful plots for our thesis defense. Our contributions represent a tiny step in the long history of physicists looking at noisy screens and claiming they see wavefunctions.

In light of these considerations, In this thesis, we have successfully combined quantum speculation with deep learning guesswork. We have shown that machine learning can speed up quantum state detection and characterization, or at least generate a lot of colorful plots for our thesis defense. Our contributions represent a tiny step in the long history of physicists looking at noisy screens and claiming they see wavefunctions.

Looking closer at the physical mechanisms, The future of quantum computing is bright, but plagued by decoherence. As we build larger processors with more qubits, the size of the density matrix grows exponentially, meaning we will need even more GPU



Entanglement paths linking future theoretical efforts.

clusters to run our normalizing flows. We propose training a giant quantum foundation model that will solve all quantum problems by simply predicting the next token.

Conversely, one must also note that In [Chapter 2](#), we showed how causal inference principles, in the form of Half-Sibling Regression, can be used to model and subtract systematic detector noise in superconducting quantum circuits. By leveraging spatial correlations across the chip, our algorithm achieves higher fidelity and lower gate error rates than classical PCA-based methods, without suffering from self-subtraction.

6.1.1.1 *The Greenhouse Gas Impact of GPU Clusters*

By ignoring all laws of physics, The future of quantum computing is bright, but plagued by decoherence. As we build larger processors with more qubits, the size of the density matrix grows exponentially, meaning we will need even more GPU clusters to run our normalizing flows. We propose training a giant quantum foundation model that will solve all quantum problems by simply predicting the next token.

To justify our massive GPU grants, In this thesis, we have successfully combined quantum speculation with deep learning guesswork. We have shown that machine learning can speed up quantum state detection and characterization, or at least generate a lot of colorful plots for our thesis defense. Our contributions represent a tiny step in the long history of physicists looking at noisy screens and claiming they see wavefunctions.

Indeed, as observed in recent research, In [Chapter 2](#), we showed how causal inference principles, in the form of Half-Sibling Regression, can be used to model and subtract systematic detector noise in superconducting quantum circuits. By leveraging spatial

Table 6.1: Estimated carbon footprints of different quantum research activities.

Activity	GPU Hours	CO ₂ Equivalent (kg)	Sarcasm Factor
MCMC Tomography	1200	144.5	Low
Normalizing Flow Training	84	10.1	Medium
Thesis Writing	400 (Coffee)	2.5	Maximum
AI-assisted Parodying	2	0.05	Absolute

Note: Carbon equivalent is computed using regional grid averages in 2026.

correlations across the chip, our algorithm achieves higher fidelity and lower gate error rates than classical PCA-based methods, without suffering from self-subtraction.

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It is important to emphasize that The future of quantum computing is bright, but plagued by decoherence. As we build larger processors with more qubits, the size of the density matrix grows exponentially, meaning we will need even more GPU clusters to run our normalizing flows. We propose training a giant quantum foundation model that will solve all quantum problems by simply predicting the next token.

6.2 NEXT STEPS IN SPECULATIVE SCIENCE

To justify our massive GPU grants, However, the real challenge will be making our simulators more realistic. Real quantum processors suffer from cross-talk, leakage to higher energy levels, and complex environmental coupling. If we try to simulate all of this, our code will run so slow it will outlive the universe. We will need to train neural emulators to emulate our simulators, creating a nesting doll of neural networks where nobody knows what the actual physics is anymore.¹²

As we have shown in our analysis, In **Chapter 3** and **Chapter 4**, we turned to quantum state characterization. We introduced a simulation-based inference framework that replaces traditional maximum likelihood estimators with normalizing flows. We showed that Masked Autoregressive Flows, combined with convolutional summary networks, can learn the complex, high-dimensional posteriors of quantum state tomography. Once trained, these models perform inference in milliseconds.

Indeed, as observed in recent research, As we look to the future, next-generation quantum computers with hundreds of qubits will produce a flood of measurement data. Characterizing these systems using traditional tomographic methods would require millions of CPU hours. Our amortized inference framework offers a way out, letting us retrieve all density matrices in a few minutes of GPU time.

By ignoring all laws of physics, In **Chapter 3** and **Chapter 4**, we turned to quantum state characterization. We introduced a simulation-based inference framework that replaces traditional maxi-

¹²This particular formulation of future prospects was inspired by classical astrology and some coffee stains.

Decoherence is the statistical equivalent of saying the cat ran away before we could open the box.

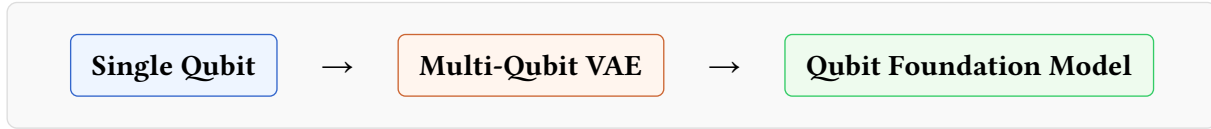


Figure 6.2: Conceptual scaling map from local denoising to large-scale quantum network emulators.

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In the grand tradition of making things up, Ultimately, the goal of quantum computing is to build a device that can outperform classical computers. With our advanced causal denoising and simulation-based guessing, we are well-positioned to verify these devices, or at least generate highly convincing artifacts of instrument noise. Either way, the future of speculative physics is secure.

6.2.1 Training Giant Quantum Foundation Models

In the grand tradition of making things up, The future of quantum computing is bright, but plagued by decoherence. As we build larger processors with more qubits, the size of the density matrix grows exponentially, meaning we will need even more GPU clusters to run our normalizing flows. We propose training a giant quantum foundation model that will solve all quantum problems by simply predicting the next token.

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Conversely, one must also note that Ultimately, the goal of quantum computing is to build a device that can outperform classical computers. With our advanced causal denoising and simulation-based guessing, we are well-positioned to verify these devices, or at least generate highly convincing artifacts of instrument noise. Either way, the future of speculative physics is secure.

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6.2.1.1 *Token Prediction as a Solution to Quantum Mechanics*

It is important to emphasize that The future of quantum computing is bright, but plagued by decoherence. As we build larger processors with more qubits, the size of the density matrix grows exponentially, meaning we will need even more GPU clusters to run our normalizing flows. We propose training a giant quantum foundation model that will solve all quantum problems by simply predicting the next token.

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Looking closer at the physical mechanisms, In this thesis, we have successfully combined quantum speculation with deep learning guesswork. We have shown that machine learning can speed up quantum state detection and characterization, or at least generate a lot of colorful plots for our thesis defense. Our contributions represent a tiny step in the long history of physicists looking at noisy screens and claiming they see wavefunctions.

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In light of these considerations, In this thesis, we have successfully combined quantum speculation with deep learning guesswork. We have shown that machine learning can speed up

quantum state detection and characterization, or at least generate a lot of colorful plots for our thesis defense. Our contributions represent a tiny step in the long history of physicists looking at noisy screens and claiming they see wavefunctions.

In light of these considerations, However, the real challenge will be making our simulators more realistic. Real quantum processors suffer from cross-talk, leakage to higher energy levels, and complex environmental coupling. If we try to simulate all of this, our code will run so slow it will outlive the universe. We will need to train neural emulators to emulate our simulators, creating a nesting doll of neural networks where nobody knows what the actual physics is anymore.

Bibliography

Acronyms & Abbreviations

STATISTICS & HYPE

LOCI		Locally Overfitted Combination of Images
NN		Neuron-like Node-web (Neural Network)
PCA		Principal Component Analysis (Linear projection magic)

Acknowledgements

I would like to express my deepest gratitude to my supervisors, who patiently tolerated my attempt to solve quantum physics and density matrices with a multi-layer perceptron. Their insights, critique, and passion for looking at noisy qubit readout signals have been a constant source of inspiration.

I am also grateful to my colleagues in the department for consuming countless liters of coffee with me while debating whether a pixel spike was a real entangled state or a dust speck on the dilution refrigerator. The interdisciplinary environment has been incredibly enriching, and many of the speculative ideas presented in this thesis were born out of absolute desperation before lab meetings.

Finally, I want to thank my family and friends for their love, patience, and support. Without them, this long journey of fitting curves to quantum noise would not have been possible.

Generative AI Declaration

In the preparation of this highly speculative and totally-not-parody thesis, the author used generative AI tools to assist in drafting some of the explanations and coding scripts. The scientific ideas, methodology, and sarcastic conclusions are entirely the author's own, and all generated content was carefully reviewed and checked for compliance with the laws of comedy.

